

While the median tells you about the center of a data set, you can use a box plot to answer questions about an entire set of data.

median = 2nd quartile

quartile = $\frac{1}{4}$ of the data set

Hide

Ms. Rivera recorded the following test scores. Find the upper and lower quartiles of the data.

55 63 68 76 79 80 80 81 83 93 94 98

$$\frac{80+80}{2} = 80$$

Step 1

Find the median. Make sure the data is arranged in order from least to greatest.

Of the 12 values, the two middle values are 80. The median is 80.

Step 2

Split the data in half at the median. Find the median of the lower half.

55 63 68 76 79 80

$$68 + 76 = 144$$

$$144 \div 2 = 72$$

The median of the lower half is 72.

Step 3

Find the median of the upper half.

80 81 83 93 94 98

$$83 + 93 = 176$$

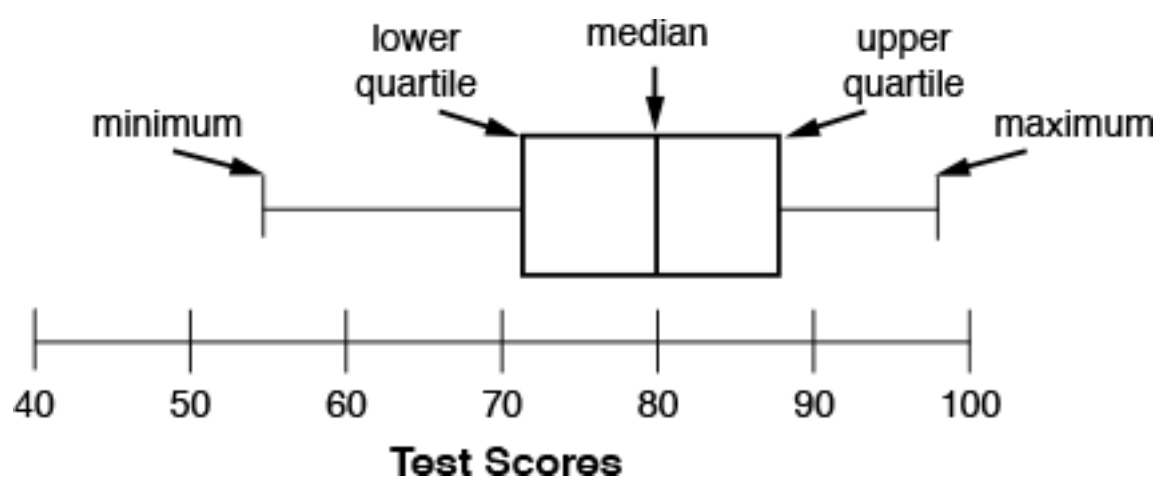
$$176 \div 2 = 88$$

The median of the upper half is 88.

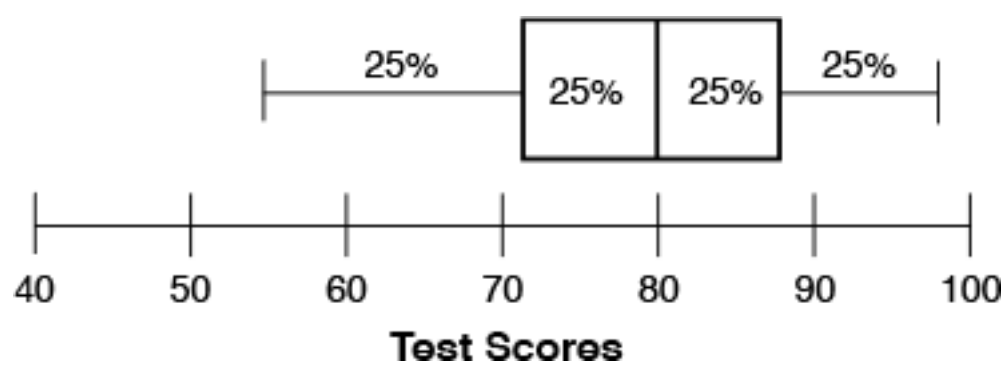
The lower quartile is 72. The upper quartile is 88.

A box plot is drawn on a scale. Draw a box that starts at the lower quartile and ends at the upper quartile. Draw a vertical line inside the box at the median of the entire data set. Then draw lines extending to the minimum and the maximum. These lines are called **whiskers**. This kind of chart is also called a **box-and-whisker plot**.

Look at the box plot of the test scores.



Each section of the plot represents 25% of the test scores.



Students in the *bottom* 25% of Ms. Rivera’s class scored between 55 and 72. Students in the *top half* of her class scored between 80 and 98.

Rate Your Confidence

^ High

= Medium

Low

Category

Lesson 2: Interpreting Graphs

 **Related Questions:**
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